

Solution of exercise 8.4.:

Since $[z]_n = [z + n]_n$ and $\gcd(n, z) = \gcd(n, z + n)$, we can w.l.o.g assume that z is positive.

Claim 1 : $\text{Ord}([z]_n) \cdot z$ is a common multiple of n and z . From the definition of the order of a group element we have

$$\text{Ord}([z]_n) \cdot [z]_n = [0]_n.$$

This already implies that

$$\text{Ord}([z]_n) \cdot z = q \cdot n \quad \text{for some } q \in \mathbb{N}$$

Consequently, $\text{Ord}([z]_n) \cdot z$ is a common multiple of z and n .

Claim 2 : $\text{Ord}([z]_n) \cdot z$ is the least common multiple of n and z . Assume that there exists a smaller common multiple of z and n . Then we have some $p_1, p_2 \in \mathbb{N}$ such that:

$$p_1 \cdot n = p_2 \cdot z < \text{Ord}([z]_n) \cdot z$$

From this we can conclude that $p_2 \cdot [z]_n = [0]_n$ and $p_2 < \text{Ord}([z]_n)$. But this contradicts the definition of Ord . So $\text{Ord}([z]_n) \cdot z$ must be the least common multiple of n and z .

Now we have:

$$\text{Ord}([z]_n) \cdot z = \text{lcm}(n, z) = \frac{n \cdot z}{\gcd(n, z)}$$

Multiplying both sides with z^{-1} yields the statement we wanted to prove.